RAG Learning

C5 — Chemical Analysis

OCR GCSE Chemistry B (Twenty First Century Science, J258) — All papers (J258/01–04, Breadth and Depth in chemistry, Foundation and Higher) · (HT) = Higher tier only

Chemical analysis answers two questions: **what** is in a sample (**qualitative analysis**) and **how much** of each substance is there (**quantitative analysis**). This chapter covers purity and separation (C5.1), identifying ions and elements (C5.2), calculating amounts in reactions (C5.3) and measuring amounts in solution by titration (C5.4).

C5.1 How are chemicals separated and tested for purity?

Formulations

> A **formulation** is a useful mixture made by combining several pure substances in **carefully measured, definite proportions** so the product does its job. Each component has a purpose.

Examples: medicines (active drug + fillers + binders + coatings), paints (pigment + binder + solvent + additives), fuels, cleaning products, cosmetics, fertilisers and alloys. Changing the proportions changes how the product behaves, so manufacturers must control the amount of every ingredient — and keep out impurities.

What does "pure" mean?

Scientific meaning	Everyday meaning

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A **pure substance** contains only **one element or one compound** — nothing else mixed in.	"Pure" means natural, unprocessed or with nothing *added*, e.g. "pure orange juice" or "pure mountain water".
Pure water is only H_2O molecules.	"Pure" orange juice is a mixture of water, sugars, acids, vitamins… so it is **not** pure in the scientific sense.

Ideas about Science — the particle model: in a pure substance every particle (atom, molecule or formula unit) is the same. In a mixture there are two or more kinds of particle. Different particles get in the way of the regular arrangement of a solid, which is why a mixture melts at a lower temperature and over a range.

Using melting points to test purity

- A **pure** substance melts (and boils) at a **sharp, fixed temperature** that matches the reference (data book) value.
- An **impure** substance melts **at a lower temperature** and **over a range** of temperatures. (Impurities also raise the boiling point.)

Sample (illustrative data)	Melting point measured	Conclusion (reference value 136 °C)
Aspirin, sample A	135.5–136 °C	Sharp, matches reference → pure
Aspirin, sample B	124–131 °C	Lower and over a 7 °C range → impure; needs further purification (e.g. recrystallisation)

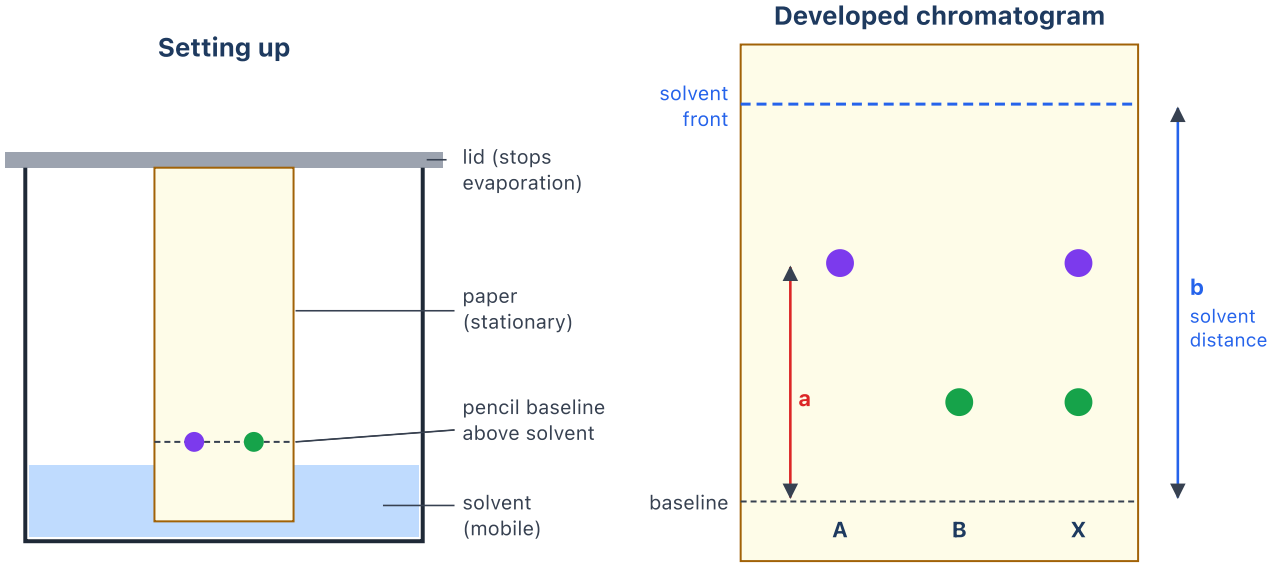
Chromatography

> Chromatography separates the substances in a mixture using two phases: a **stationary phase** that does not move (e.g. the paper, or the water held in it) and a **mobile phase** that moves through it (the **solvent**). Separation depends on how each substance is **distributed between the two phases**.

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- A substance that is **more soluble in the mobile phase** (less attracted to the stationary phase) spends more time moving and travels **further**.
- A substance **more strongly attracted to the stationary phase** travels a shorter distance.
- A **pure** substance gives **one spot** (in every solvent tried); a mixture gives two or more spots.

Paper Chromatography and R_f Values



R_f = a ÷ b. A: 120 ÷ 200 = 0.60; B: 50 ÷ 200 = 0.25 (illustrative scale units)

Unknown X has spots matching A and B (same R_f), so X is a mixture of A and B.

R_f = distance moved by the substance (spot) ÷ distance moved by the solvent front

both measured from the baseline; R_f has no units and is always between 0 and 1

Worked Example: A spot moves 3.6 cm while the solvent front moves 8.0 cm. Find R_f, and identify it using reference values: caffeine 0.45, glucose 0.30 (illustrative values for this solvent).

R_f = 3.6 ÷ 8.0 = **0.45** → matches caffeine. To be confident, run the reference and the unknown side by side on the same paper, and repeat in a second solvent: they should match in both.

Choosing a chromatography method (PAG4)

- **Paper chromatography** — spots placed on a **pencil** baseline (ink would itself run), baseline **above** the solvent level (otherwise the sample dissolves into the solvent in the beaker), lid on to keep the atmosphere saturated with solvent.
- **Aqueous and non-aqueous solvents** — water works for water-soluble dyes (e.g. food colourings). Substances **insoluble in water** (e.g. permanent-marker inks, chlorophyll) need a non-aqueous solvent such as ethanol or propanone. R_f values depend on the solvent, so a reference R_f is only valid for the same solvent.
- **Locating agents** — colourless substances (e.g. amino acids, sugars) do not show up. After drying, the chromatogram is sprayed with a locating agent (e.g. ninhydrin for amino acids) or put under UV light, which makes the spots visible.
- **Distinguishing pure from impure** — a pure substance gives one spot that matches the reference; extra spots reveal impurities. Using two different solvents gives stronger evidence, because two substances could by chance have the same R_f in one solvent.

Separation techniques (PAG3, PAG7)

Separation methods exploit a **difference in a property** of the components: state, solubility in a solvent, or boiling point. Reactions in the lab and industry often give the wanted product mixed with by-products, waste and unreacted reactants, so separation is part of almost every process.

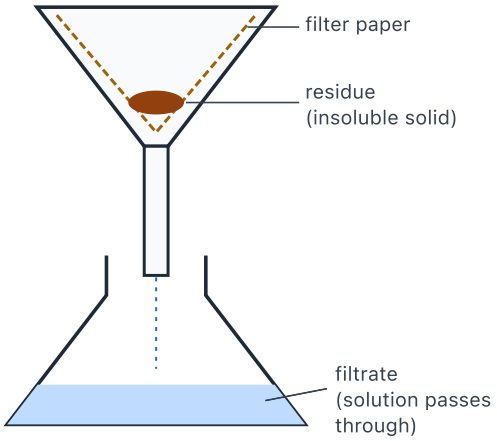
Technique	Separates…	Property used	Example
Filtration	An insoluble solid from a liquid/solution	Solid particles too big to pass through filter paper	Sand from salty water; excess copper(II) oxide from copper sulfate solution
Crystallisation	A dissolved solid (solute) from its solution	Solute less soluble when the solution is concentrated and cooled	Copper sulfate crystals from solution
Simple distillation	A liquid (solvent) from a solution	Big difference in boiling point	Pure water from sea water

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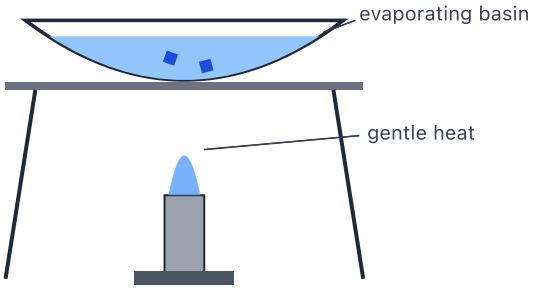
Fractional distillation	Miscible liquids with close boiling points	Different boiling points	Ethanol from water; fractions of crude oil (C3.4)
Chromatography	Small amounts of soluble substances, e.g. dyes	Distribution between mobile and stationary phases	Dyes in an ink

Filtration and Crystallisation

Filtration



Crystallisation



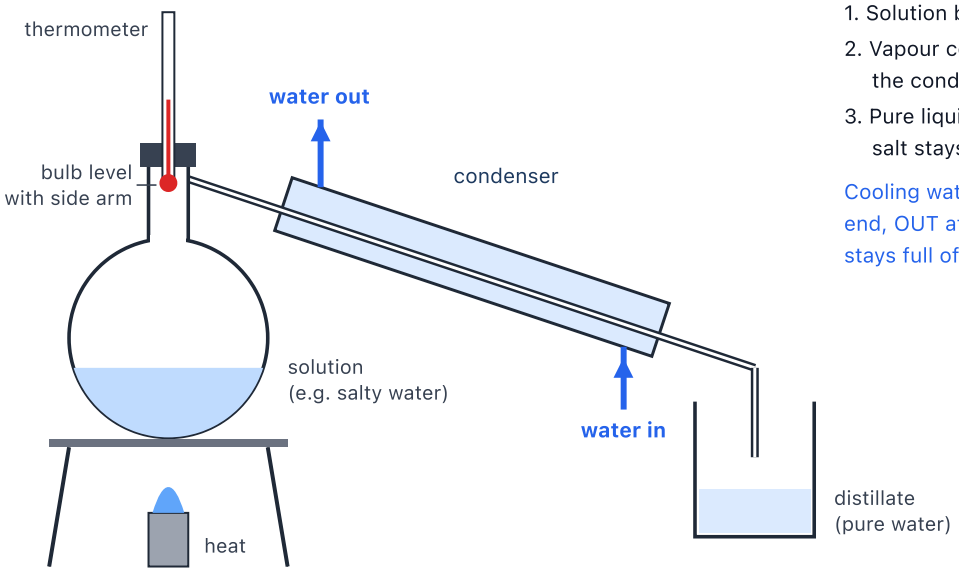
Heat until crystals first appear (point of crystallisation), then leave to cool; filter off and dry the crystals.

Distillation

Simple distillation: the solution is heated; the liquid with the lower boiling point evaporates, the vapour passes into a water-cooled **condenser**, condenses and is collected as the **distillate**. Dissolved solids stay in the flask. The thermometer bulb is level with the side arm so it reads the temperature of the vapour leaving — a steady reading at the known boiling point shows a pure liquid is distilling.

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Simple Distillation



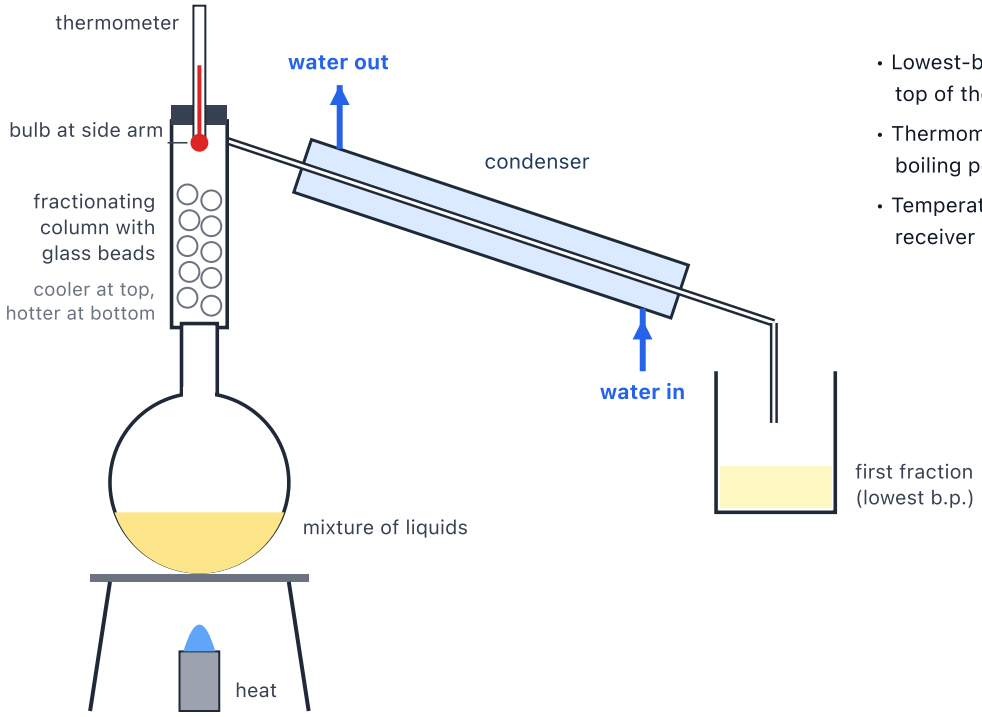
1. Solution boils; water vapour rises.
2. Vapour cools and condenses in the condenser.
3. Pure liquid drips into the beaker; salt stays in the flask.

Cooling water flows IN at the lower end, OUT at the top, so the jacket stays full of cold water.

Fractional distillation separates **mixtures of liquids** whose boiling points are close. A **fractionating column** packed with glass beads sits between the flask and the condenser. The column is hottest at the bottom and coolest at the top. Vapours condense and re-evaporate many times on the beads; the liquid with the **lowest boiling point** reaches the top first and distils over while the thermometer reading stays steady. When the temperature rises to the next boiling point, the receiver is changed to collect the next **fraction**. Crude oil is separated this way on an industrial scale (C3.4).

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Fractional Distillation (e.g. ethanol from water)



- Lowest-boiling liquid reaches the top of the column first.
- Thermometer stays steady at its boiling point while it distils.
- Temperature rises → change the receiver for the next fraction.

Suggesting a purification technique

- Insoluble solid + liquid → **filtration**.
- Want the dissolved solid → **crystallisation** (evaporating to dryness can decompose some salts, e.g. hydrated salts lose water).
- Want the solvent from a solution → **simple distillation**.
- Two or more miscible liquids → **fractional distillation**.
- Two solids, one soluble in a solvent → dissolve, **filter**, then **crystallise** the filtrate.
- Analysing or separating tiny amounts of coloured/soluble substances → **chromatography**.

Separations are **rarely completely successful**: products often go through several stages, or the same process repeatedly (e.g. recrystallising), to reach an acceptable purity. Purity is then checked by melting point or chromatography.

> **Exam Tip:** For "suggest a method" questions, name the technique *and* give the property it relies on, using the data you are given: "fractional distillation, because the boiling points (78 °C and 100 °C) are different".

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C5.2 How do chemists find the composition of unknown samples?

Representative sampling

The small sample analysed must **represent the whole bulk** of the material (a soil field, a batch of product, a river). Chemists take several samples from different places/times and mix or analyse them separately so that variations in the bulk are detected and the result is not biased by one unusual spot. **Ideas about Science (IaS1):** plan the equipment, techniques and strategy for a qualitative analysis — e.g. sample, flame test for the cation, then add reagents to test for anions.

Flame tests (PAG5)

Technique: clean a nichrome (or platinum) wire loop by dipping it in dilute hydrochloric acid and holding it in a hot (blue, roaring) Bunsen flame until it gives no colour. Dip the loop into the solid sample (or solution), hold it at the edge of the hot blue flame and observe the colour. Clean the loop between samples to avoid contamination.

Metal ion	Flame colour
Lithium, Li^+	Crimson (red)
Sodium, Na^+	Yellow (orange-yellow)
Potassium, K^+	Lilac
Calcium, Ca^{2+}	Orange-red (brick red)
Copper, Cu^{2+}	Green (blue-green)

Limitation: a mixture of ions can mask colours (sodium's strong yellow hides others), and small amounts may not be seen.

Precipitation tests for metal ions with dilute sodium hydroxide (PAG5)

Add a few drops of dilute NaOH(aq) to a solution of the sample, then add excess. Insoluble metal hydroxides precipitate.

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Ion	Result with a few drops of NaOH(aq)	With excess NaOH	Ionic equation
Calcium, Ca^{2+}	White precipitate	Remains (insoluble)	$Ca^{2+}(aq) + 2OH^-(aq) \rightarrow Ca(OH)_2(s)$
Zinc, Zn^{2+}	White precipitate	**Dissolves** → colourless solution	$Zn^{2+}(aq) + 2OH^-(aq) \rightarrow Zn(OH)_2(s)$
Copper(II), Cu^{2+}	Blue precipitate	Remains	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$
Iron(II), Fe^{2+}	Green precipitate (turns orange-brown at the surface on standing in air)	Remains	$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$
Iron(III), Fe^{3+}	Orange-brown precipitate	Remains	$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$

Calcium and zinc both give white precipitates: tell them apart with excess NaOH (zinc hydroxide dissolves) or a flame test (calcium is orange-red).

Tests for anions (PAG5)

Ion	Test	Positive result	Equation
Carbonate, CO_3^{2-}	Add dilute acid; bubble gas through limewater	Fizzing; gas turns limewater milky (CO_2)	$CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$
Chloride, Cl^-	Acidify with dilute **nitric acid**, then add silver nitrate solution	**White** precipitate (AgCl)	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
Bromide, Br^-		**Cream** precipitate (AgBr)	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$

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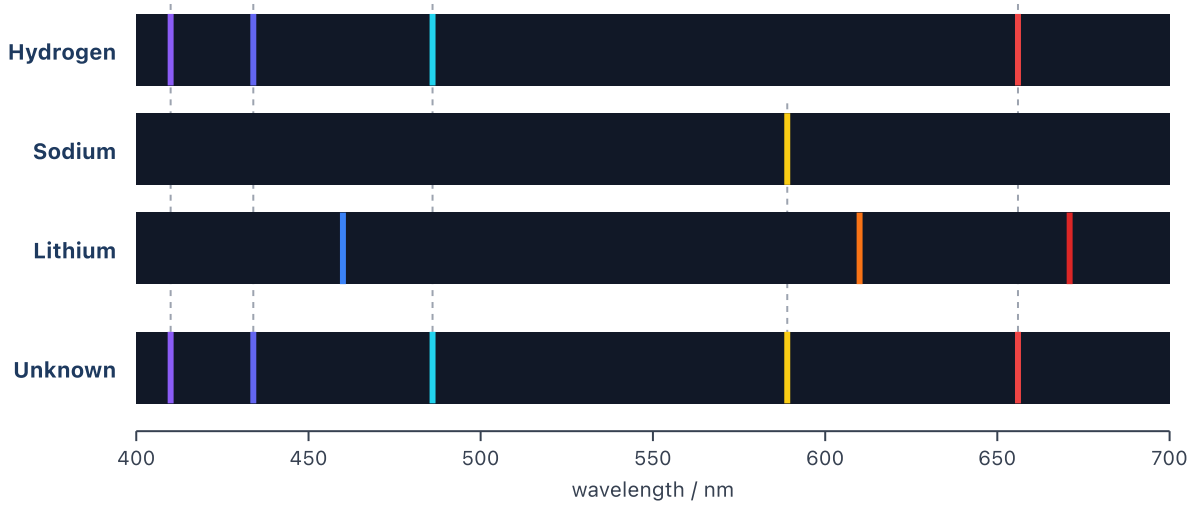
Iodide, I^-		**Yellow** precipitate (AgI)	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$
Sulfate, SO_4^{2-}	Acidify with dilute hydrochloric acid, then add barium chloride solution (or acidify with nitric acid and add barium nitrate)	**White** precipitate ($BaSO_4$)	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

Exam Tip: Why acidify first? The acid removes carbonate ions, which would otherwise also form a precipitate and give a false positive. Use **nitric** acid (not hydrochloric) before silver nitrate, because HCl adds chloride ions; never use sulfuric acid before the barium test, because it adds sulfate ions.

Instrumental analysis: emission spectroscopy

A sample is heated strongly (e.g. in a flame or plasma) so its atoms emit light. The light is split into a **line spectrum**. Each element gives its **own unique pattern of lines** at particular wavelengths, like a fingerprint. Matching the lines of an unknown to **reference spectra** identifies the elements present. A mixture shows **all** the lines of every element in it. Uses: identifying elements in stars and checking the composition of alloys such as steel.

Interpreting Emission Spectra (simplified line positions)



Every hydrogen and sodium line appears in the unknown, and no lithium lines → it contains H and Na only.

Instrumental vs laboratory (wet) methods

Instrumental methods (e.g. emission spectroscopy, gas chromatography)	Standard laboratory tests (flame tests, precipitates)
Sensitive — detect tiny amounts of a substance	Need larger samples; faint results easily missed
Accurate — results match the true value; can identify substances in mixtures and measure how much	Rely on judging colours by eye; mixtures can mask each other
Rapid — fast, automated, many samples; data recorded automatically	Slower, one test at a time
Expensive, need trained operators, not widely available	Cheap glassware and reagents, available in any lab

> **Exam Tip:** When interpreting a spectrum or chart (M4a), go line by line: say which reference lines match which lines in the unknown, then conclude. A single unmatched line means another element is present.

C5.3 How are the amounts of substances in reactions calculated?

Conservation of mass

> **Law of conservation of mass:** in a chemical reaction no atoms are created or destroyed — they are only rearranged. So the **total mass of products = total mass of reactants**.

In a **non-enclosed (open) system** the mass can appear to change, because a gas can enter or leave:

- **Mass decreases** when a gas product escapes, e.g. $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$. The CO_2 particles leave the flask into the air.
- **Mass increases** when a reactant comes from the air, e.g. burning magnesium: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$. Oxygen atoms from the air are now combined in the solid.

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In a sealed container the mass stays exactly the same, because all the particles are still there.

Relative formula mass, M_r

Add the relative atomic masses (A_r, from the Periodic Table) of all the atoms in the formula.

Worked Example: M_r of calcium carbonate, $CaCO_3$ = 40 + 12 + (3 × 16) = **100**.
M_r of calcium hydroxide, $Ca(OH)_2$ = 40 + 2 × (16 + 1) = **74**.
In $2H_2 + O_2 \rightarrow 2H_2O$: mass of reactants = 2 × 2 + 32 = 36; mass of products = 2 × 18 = 36 — mass is conserved.

The mole and the Avogadro constant (HT)

The **Avogadro constant** is **6.02×10^{23}** per mole. One **mole** of a substance contains 6.02×10^{23} particles (atoms, molecules, ions or formula units). The mass of one mole is its M_r in grams. (OCR accepts either definition: the number of atoms in exactly 12 g of carbon-12, or exactly 6.02×10^{23} particles.)

number of moles = mass of substance (g) ÷ relative formula mass

rearranged: $\text{mass} = \text{moles} \times M_r$ $\quad$ $\text{number of particles} = \text{moles} \times 6.02 \times 10^{23}$

Worked Example: (HT) How many molecules are in 9.0 g of water (M_r 18)? How many atoms?
moles = 9.0 ÷ 18 = 0.50 mol. Molecules = $0.50 \times 6.02 \times 10^{23}$ = **3.01×10^{23}**. Each molecule has 3 atoms, so atoms = **9.03×10^{23}**.

Deducing an equation from reacting masses (HT)

Worked Example: (HT) 4.8 g of magnesium reacts with 3.2 g of oxygen, O_2, to give 8.0 g of magnesium oxide. Deduce the balanced equation (A_r: Mg 24, O 16).

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Mg: 4.8 ÷ 24 = 0.20 mol; O_2: 3.2 ÷ 32 = 0.10 mol; MgO: 8.0 ÷ 40 = 0.20 mol.
Divide by the smallest (0.10): ratio 2 : 1 : 2 → **$2Mg + O_2 \rightarrow 2MgO$**.

Limiting reactants (HT)

The **limiting reactant** is the one that is completely used up first; it stops the reaction and fixes the amount of product. The other reactant is **in excess** (some is left over). Doubling the limiting reactant doubles the product; adding more of the excess reactant makes no difference.

Worked Example: (HT) 2.4 g Mg is added to 0.10 mol HCl. $Mg + 2HCl \rightarrow MgCl_2 + H_2$. Which is limiting?
Mg = 2.4 ÷ 24 = 0.10 mol, which needs 0.20 mol HCl. Only 0.10 mol HCl is present, so **HCl is limiting** and Mg is in excess. H_2 formed = 0.10 ÷ 2 = **0.050 mol**.

Reacting-mass calculations (HT)

Steps: (1) moles of the known substance = mass ÷ M_r; (2) use the **ratio** in the balanced equation; (3) mass of the unknown = moles × M_r. This lets industry use reactants in the correct proportions so processes run efficiently with little waste.

Worked Example: (HT) What mass of calcium oxide is made by heating 50 g of calcium carbonate? $CaCO_3 \rightarrow CaO + CO_2$
moles $CaCO_3$ = 50 ÷ 100 = 0.50 mol; ratio 1 : 1 so 0.50 mol CaO; mass = 0.50 × 56 = **28 g**.

Theoretical yield, actual yield and percentage yield

The **theoretical yield** is the maximum mass of product calculated from the equation. The **actual yield** is the mass you really obtain.

$$\text{percentage yield} = (\text{actual yield} \div \text{theoretical yield}) \times 100$$

Worked Example: In the example above only 22.4 g of CaO was collected. Percentage yield

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= 22.4 ÷ 28 × 100 = **80%**.

Without moles (all tiers) you can scale masses by ratio: if 100 g $CaCO_3$ gives 56 g CaO, then 50 g gives 56 × 50/100 = 28 g.

Reasons for low yields:

- The reaction is **reversible** and does not go to completion (C6.3).
- **Impure reactants** — less of the actual reactant than the mass suggests.
- Product **lost during the procedure**: left on filter paper or in glassware, lost when transferring, some stays dissolved in the solution during crystallisation.
- **Side reactions** make unwanted products.

Ideas about Science (IaS1, IaS2): use data to make quantitative predictions of yield, then compare them with actual yields — e.g. preparing a salt or an organic compound and comparing the mass made with the theoretical yield, or predicting the volume of gas made when an acid reacts and testing it. Information about actual yields is used to improve the procedure.

Moles and gas volumes (HT)

One mole of **any gas** occupies **24 dm³** (24 000 cm³) at room temperature and pressure (RTP).

number of moles of gas = volume of gas (dm³) ÷ 24 dm³ $\quad \text{volume} = \text{moles} \times 24$

- For reactions where **everything is a gas**, the volume ratio equals the mole ratio in the equation: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, so 30 dm³ of H_2 makes 20 dm³ of NH_3 (with enough N_2).
- For mixed states, combine mass and volume steps.

Worked Example: (HT) What volume of CO_2 (at RTP) is made when 10 g of $CaCO_3$ reacts with excess hydrochloric acid?
$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. moles $CaCO_3$ = 10 ÷ 100 = 0.10 mol → 0.10 mol CO_2 → volume = 0.10 × 24 = **2.4 dm³** (2400 cm³).

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Exam Tip: Rearranging formulae (M3b/M3c) is examined directly. Write the formula, rearrange, substitute with units, and give the answer to a sensible number of significant figures. Use standard form for Avogadro answers (e.g. 3.01×10^{23}, not 301 000 000 000 000 000 000 000).

C5.4 How are the amounts of chemicals in solution measured?

Qualitative vs quantitative analysis

Qualitative analysis	Quantitative analysis
Identifies **which** substances/ions are present (e.g. flame tests, precipitate tests, chromatography to identify a dye)	Measures **how much** of a substance is present (e.g. titration to find a concentration, measuring a mass or a gas volume)

Concentration of solutions (HT)

concentration (g/dm³) = mass of solute (g) ÷ volume of solution (dm³)

concentration (mol/dm³) = number of moles of solute ÷ volume of solution (dm³)

1 dm³ = 1000 cm³, so volume in dm³ = volume in cm³ ÷ 1000

More solute in the same volume, or the same solute in a smaller volume, gives a higher concentration. Chemists usually prefer **mol/dm³** because it links directly to the reacting ratios in the equation. To convert: concentration in mol/dm³ = concentration in g/dm³ ÷ M_r.

Worked Example: (HT) 4.0 g of sodium hydroxide (M_r 40) is dissolved to make 250 cm³ of solution.
Volume = 250 ÷ 1000 = 0.250 dm³. Concentration = 4.0 ÷ 0.250 = **16 g/dm³**.
Moles = 4.0 ÷ 40 = 0.10 mol, so concentration = 0.10 ÷ 0.250 = **0.40 mol/dm³** (check: 16 ÷ 40 = 0.40).

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Making a standard solution (a solution of accurately known concentration): weigh the solid accurately, dissolve it in a beaker of distilled water, transfer to a **volumetric flask** with rinsings, make up to the graduation mark with distilled water (bottom of meniscus on the line), stopper and invert to mix.

Neutralisation

Acids form **hydrogen ions, $H^+(aq)$**, when they dissolve in water. Solutions of **alkalis** (soluble bases) contain **hydroxide ions, $OH^-(aq)$**.

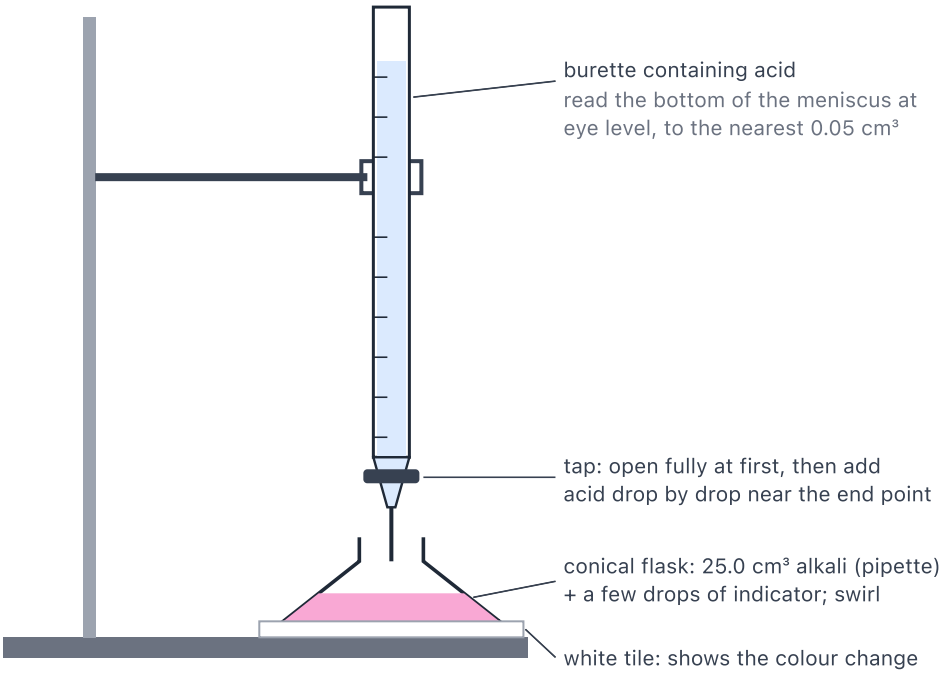
> **Neutralisation:** acid + alkali → salt + water. Every aqueous neutralisation can be generalised as the ionic equation **$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$**.

Acid + alkali	Balanced equation	Salt
Hydrochloric acid + sodium hydroxide	$HCl + NaOH \rightarrow NaCl + H_2O$	sodium chloride
Nitric acid + potassium hydroxide	$HNO_3 + KOH \rightarrow KNO_3 + H_2O$	potassium nitrate
Sulfuric acid + sodium hydroxide	$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$	sodium sulfate
Hydrochloric acid + calcium hydroxide	$2HCl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O$	calcium chloride
Nitric acid + calcium hydroxide	$2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$	calcium nitrate

Titration procedure (PAG6)

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Acid–Alkali Titration



1. Rinse the pipette with the alkali and use a **pipette filler** to transfer exactly **25.0 cm³** of alkali into a conical flask.
2. Add a few drops of a suitable indicator (e.g. **phenolphthalein**: pink in alkali, colourless in acid; or **methyl orange**: yellow in alkali, red in acid). Universal indicator is not used — its gradual colour change gives no sharp end point.
3. Rinse the burette with the acid, fill it (using a funnel, then remove the funnel) and make sure the jet below the tap is full. Record the starting reading.
4. Place the flask on a **white tile**. Add acid, swirling, until the indicator just changes colour (the **end point**). This first, quick run is the **rough titration**.
5. Repeat accurately: add acid quickly to within about 1 cm³ of the rough value, then **drop by drop**, swirling, until one drop changes the colour permanently.
6. Repeat until you have **concordant** titres (within 0.10 cm³ of each other). Calculate the **mean of the concordant titres**, ignoring the rough.

Evaluating titration data

Run (illustrative data)	Rough	1	2	3
Final reading / cm³	25.40	24.70	49.25	24.65

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Initial reading / cm^3	0.00	0.00	24.60	0.00
Titre / cm^3	25.40	24.70	24.65	24.65

The rough is discarded. Titres 1–3 have a **range** of 24.70 – 24.65 = 0.05 cm^3, so they are concordant. Mean = (24.70 + 24.65 + 24.65) ÷ 3 = **24.67 cm^3**.

- **Precise / repeatable:** repeat titres close together (small range).
- **Accurate:** close to the true value — achieved with calibrated burette and pipette, standard solution of known concentration, reading the meniscus at eye level and adding dropwise at the end point.
- **Valid:** the method measures what it claims to — same volume of alkali each time, same indicator, only the acid volume varies.
- **Risk:** wear eye protection (acids and alkalis are irritant/corrosive); use a pipette filler, never mouth-pipette; fill the burette below eye level.

Ideas about Science (IaS1, IaS2): justify each step of the technique in terms of the precision, accuracy and validity of the data and minimising risk; use the range and mean to process titration results.

Titration calculations (HT)

The volume of a solution of **known concentration** that reacts exactly with another substance lets you find the other's concentration (or volume) using the reacting ratio:

1. moles of the known solution = concentration × volume (dm^3)
2. use the equation ratio to find moles of the unknown
3. concentration of the unknown = moles ÷ volume (dm^3)

Worked Example: (HT) 25.0 cm^3 of 0.100 mol/dm^3 NaOH is neutralised by 20.0 cm^3 of hydrochloric acid. Find the concentration of the acid.
moles NaOH = 0.100 × 25.0/1000 = 0.00250 mol. HCl + NaOH → NaCl + H_2O, ratio 1 : 1 → 0.00250 mol HCl.
[HCl] = 0.00250 ÷ (20.0/1000) = **0.125 mol/dm^3**.

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Worked Example: (HT) 25.0 cm^3 of 0.200 mol/dm^3 NaOH reacts exactly with 25.0 cm^3 of sulfuric acid. $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$.
moles NaOH = 0.200 × 0.0250 = 0.00500 mol; ratio 1 H_2SO_4 : 2 NaOH → 0.00250 mol H_2SO_4.
$[H_2SO_4]$ = 0.00250 ÷ 0.0250 = **0.100 mol/dm^3**.

Exam Tip: Always convert cm^3 to dm^3 (÷ 1000) before using mol/dm^3, and check the ratio in the equation — sulfuric acid is the classic trap because it reacts 1 : 2 with NaOH.

C5 Summary & Key Equations

- Pure (science) = one substance only; pure substances have sharp melting points.
- R_f = distance moved by spot ÷ distance moved by solvent front
- Filtration, crystallisation, simple and fractional distillation each exploit a different property.
- Flame tests: Li crimson, Na yellow, K lilac, Ca orange-red, Cu green.
- NaOH: Ca^{2+} white, Zn^{2+} white (dissolves in excess), Cu^{2+} blue, Fe^{2+} green, Fe^{3+} orange-brown.
- Ag^+ test: Cl^- white, Br^- cream, I^- yellow; Ba^{2+} test: SO_4^{2-} white; carbonate fizzes, CO_2 turns limewater milky.
- (HT) moles = mass ÷ M_r; particles = moles × 6.02 × 10^{23}; gas moles = volume (dm^3) ÷ 24
- percentage yield = actual ÷ theoretical × 100
- (HT) concentration (g/dm^3) = mass ÷ volume; concentration (mol/dm^3) = moles ÷ volume
- Neutralisation: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

Exam Tip: In a 6-mark titration question, cover the apparatus (burette, pipette, indicator, tile), the method (rough, dropwise near end point, concordant repeats) and the processing (mean of concordant results) — each strand earns marks.

Spec Checklist — C5

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Spec	You should be able to...
C5.1.1	Explain that many useful materials are formulations of mixtures
C5.1.2	Explain what is meant by purity, distinguishing scientific and everyday uses of "pure"
C5.1.3	Use melting point data to distinguish pure from impure substances
C5.1.4	Recall that chromatography has a stationary and a mobile phase and separation depends on distribution between them
C5.1.5	Interpret chromatograms, including calculating R_f values
C5.1.6	Suggest chromatographic methods to distinguish pure from impure substances: paper chromatography, aqueous and non-aqueous solvents, locating agents (PAG4)
C5.1.7	Describe, explain and exemplify filtration, crystallisation, simple and fractional distillation (PAG3, PAG7)
C5.1.8	Suggest suitable purification techniques given information about the substances (PAG3, PAG7)
C5.2.1	Describe the purpose of representative sampling in qualitative analysis
C5.2.2	Interpret flame tests for lithium, sodium, potassium, calcium and copper ions
C5.2.3	Describe the technique of flame tests (PAG5)
C5.2.4	Describe tests for aqueous cations (NaOH: Ca^{2+}, Cu^{2+}, Fe^{2+}, Fe^{3+}, Zn^{2+}) and anions (CO_3^{2-}, Cl^-, Br^-, I^-, SO_4^{2-}) and identify species from results (PAG5)
C5.2.5	Interpret an emission spectroscopy result against a reference set
C5.2.6	Describe the advantages of instrumental methods (sensitivity, accuracy, speed)
C5.2.7	Interpret charts, particularly in spectroscopy
C5.3.1	Recall and use the law of conservation of mass

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C5.3.2	Explain observed mass changes in non-enclosed systems using the particle model
C5.3.3	Calculate relative formula masses separately and in a balanced equation
C5.3.4 (HT)	Recall and use the definitions of the Avogadro constant (in standard form) and of the mole
C5.3.5 (HT)	Relate mass in grams to amount in moles: moles = mass ÷ M_r
C5.3.6 (HT)	Deduce the stoichiometry of an equation from reacting masses; explain the effect of a limiting reactant
C5.3.7 (HT)	Use a balanced equation to calculate masses of reactants or products
C5.3.8	Use arithmetic, ratio, percentage and multistep calculations in quantitative chemistry
C5.3.9 (HT)	Calculate with standard form when using the Avogadro constant
C5.3.10	Change the subject of a mathematical equation
C5.3.11	Calculate the theoretical mass of a product from a given mass of reactant
C5.3.12	Calculate percentage yield from the actual yield
C5.3.13	Suggest reasons for low yields for a given procedure
C5.3.14 (HT)	Relate molar amounts of gases to volumes and calculate gas volumes using 24 dm^3 at RTP
C5.4.1	Identify the difference between qualitative and quantitative analysis
C5.4.2 (HT)	Calculate concentration in g/dm^3 = mass of solute ÷ volume
C5.4.3 (HT)	Calculate concentration in mol/dm^3 = moles of solute ÷ volume

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C5.4.4	Describe neutralisation (HCl, HNO_3, H_2SO_4 with NaOH, KOH, $Ca(OH)_2$) as acid + alkali → salt + water
C5.4.5	Recall that acids form H^+ ions in water and alkalis contain OH^- ions
C5.4.6	Recognise neutralisation as $H^+ + OH^- \rightarrow H_2O$
C5.4.7	Describe and explain a titration procedure giving precise, accurate, valid and repeatable results (PAG6)
C5.4.8	Evaluate the quality of data from titrations
C5.4.9 (HT)	Relate the volume and concentration of one solution to the volume or concentration of another that reacts completely with it

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